

# KB Pass 3 — Synthesis Report

Generated 2026-08-23 15:54 UTC from on-disk artifacts in **kb/energy\_knowledge\_base** and **kb/knowledge**. Every figure is computed from those files (canonical objects, synth\_state.json, merge\_plan.json, excluded\_from\_pass3.json, concept\_graph.json, contradictions\_review.md) — nothing projected or invented.

## 1. Canonical knowledge base

**733 canonical objects** = 24 merged (multi-item) + 709 singleton pass-throughs. Routed to typed folders:

| folder               | objects |
|----------------------|---------|
| concepts             | 291     |
| definitions          | 156     |
| formulas             | 28      |
| market_mechanics     | 131     |
| trading_implications | 23      |
| risk                 | 83      |
| examples             | 21      |
| TOTAL                | 733     |

Chapter orientation summaries generated: **34** (rewritten, not stitched).

## 2. Merge statistics

**24/24 multi-item groups merged.** Synth-judge flag rate: **8/24 = 33%** (design baseline from the 9-group pilot was ~1/3; the circuit-breaker at >50% did NOT trip). Error scars (twice-failed groups): **0**. Merge combines validated content only — formulas, source refs and the validation rollup are assembled deterministically; the LLM merges prose, then a mandatory formula-aware judge checks every merged object. Flagged objects reach the KB *flagged* (never silently dropped).

Flagged objects (synth review queue): black\_scholes, credit\_risk, hedge\_accounting, load\_forecasting, mark\_to\_market, natural\_gas\_liquids, natural\_gas\_storage, wheeling

## 3. Top-10 concepts by source breadth

Breadth = number of distinct chapters contributing to the object's source\_refs.

| #  | canonical topic               | chapters | items | which chapters |
|----|-------------------------------|----------|-------|----------------|
| 1  | heat rate                     | 5        | 7     | 1,4,5,15,18    |
| 2  | spark spread                  | 4        | 9     | 1,5,15,18      |
| 3  | hedging                       | 4        | 7     | 1,2,14,33      |
| 4  | spread option                 | 4        | 7     | 1,15,18,19     |
| 5  | mark to market                | 3        | 13    | 1,2,29         |
| 6  | wheeling                      | 3        | 12    | 1,5,19         |
| 7  | credit risk                   | 3        | 6     | 1,2,31         |
| 8  | crack spread                  | 3        | 5     | 1,6,15         |
| 9  | financial transmission rights | 3        | 4     | 1,5,25         |
| 10 | value at risk                 | 2        | 12    | 1,30           |

## 4. Contradictions — recorded UNRESOLVED

Verbatim from contradictions\_review.md. The merge records conflicting definitions/formulas with both versions + sources and does NOT resolve or average them — the KB's standard is provenance, not adjudication.

- heat rate - direction of the heat-rate/efficiency relationship

- "Lower heat rate = more efficient plant." (from: Heat rate)
- "A lower heat rate indicates a more efficient power plant." (from: Heat rate and fossil fuel plants as marginal price setters)
- "a higher heat rate figure (as used here) means the plant produces more electricity from the same amount of fuel" (from: Heat rate and plant conversion efficiency)

note: The 'Heat rate and plant conversion efficiency' item states that a higher heat rate figure means more electricity produced from the same fuel, which inverts the fuel-input-per-unit-output convention (lower = more efficient) stated by the other items. Not resolved.

- hedging - whether hedging eliminates or merely transfers risk

- "Hedging transfers risk from one trader to another rather than eliminating it." (from: Hedging (ch1, p.69))
- "Hedging means taking an offsetting position to reduce the risk of an existing position... generally used to lock in profit or eliminate risk on a position that can't practically be unwound." (from: Hedging (ch2, p.110-113))

note: One item frames hedging strictly as a risk transfer between counterparties (risk isn't destroyed, just moved), while another describes it as reducing or eliminating risk on the hedger's own position. Not resolved here.

- liquefied natural gas - density/volume reduction factor upon liquefaction

- "Liquid form is roughly 600 times denser than gaseous form." (from: Liquefied natural gas (LNG) and ocean transport)
- "As a liquid, natural gas occupies only about 1/610 of its gaseous volume." (from: Liquefied Natural Gas (LNG) Process)

note: Both describe the same liquefaction density effect but cite slightly different figures (600x vs 610x); not resolved/averaged here.

- implied heat rate - formula basis

- "Market Implied Heat Rate = Power Price / Fuel Price (current power price over current fuel price, generic fuel)" (from: Market implied heat rate)
- "Implied Heat Rate = Average Power Price / Average Natural Gas Price, computed in a given market on a given day" (from: Implied market heat rate)

note: One item defines the ratio using current/point-in-time power and (generic) fuel prices; the other specifies daily-average power and natural-gas prices for a given market. Not resolved into a single formula since the items differ on both the price convention (current vs. daily average) and fuel specificity (generic fuel vs. natural gas).

4 contradictions, all UNRESOLVED.

## 5. Concept graph

**732 nodes, 280 edges.** (nodes < objects because 1 object(s) share a canonical\_topic label, which collapses to one node) Every edge is derived ONLY from an object's stated relationship text that names another canonical concept — no edges are invented.

## 6. Reconciliation

Trusted items in = **857**; contributing items across all canonical objects = **857** → **nothing lost**. Trusted 857 + excluded 54 = **911** (the full Pass-2 population). The trusted/excluded split is recomputed from each item's `_validation`, so Pass 3 re-runs cleanly when the review queue is cleared.

## 7. Excluded items (not deleted)

**54 items excluded** from synthesis: 47 UNSUPPORTED + 7 UNCLEAR. They are listed in `excluded_from_pass3.json` (file+idx+topic+chapter+verdict), NOT removed — re-running Pass 3 after the Pass-2 review queue is cleared folds any newly-trusted item back in.

## 8. Honest scope — what is and isn't done

**Synthesis is complete.** But TWO review queues are OPEN and awaiting human triage: (a) the Pass-2 validation queue — **54 items** (UNSUPPORTED/UNCLEAR) never entered synthesis; (b) the synth queue — **8 flagged objects** whose merged prose the judge could not fully trace to their source items. Neither has been triaged by a human.

Retrieval, embeddings, vector DB and evaluation (Pass 5-6) are NOT built. The Co-Pilot is NOT wired to this KB — **no synthesized knowledge reaches the live app yet**. This report describes an offline, auditable knowledge base, not a shipped feature.